KUSH MEHTA

Curriculum Vitae

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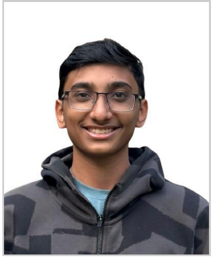

Final-year Electronics and Communication Engineering student at Nirma University, applying for MS study beginning Fall 2027. Research interests span error-correcting codes and their hardware realization, computer architecture, and secure embedded systems. Recent work includes a real-time Galileo HAS decoder built at SAC/ISRO, a synthesizable GF(2^8) arithmetic core verified against that decoder, and a co-authored paper on evaluation hazards in learned frequency-hopping receivers.

EDUCATION

Institute of Technology, Nirma University — Ahmedabad, India
B.Tech, Electronics & Communication Engineering — Minor in Data Science | CGPA 8.59 / 10 — Aug 2023 – May 2027

- **Core coursework:** Digital Design, VLSI Design, Analog CMOS Integrated Circuits, Computer Architecture, Embedded Systems, Microcontrollers & Interfacing, Signals & Systems, Digital Signal Processing, Image Processing, Communication Systems, Computer Programming, Scripting Languages, Programming for Scientific Computing
- **Data Science minor:** Machine Learning, Statistics, Data Analysis & Visualization, Analytics of IoT, Big Data Systems

RESEARCH & PROFESSIONAL EXPERIENCE

Research Intern — Navigation Receiver Division — May – Jul 2026
Space Applications Centre (SAC), ISRO · Real-Time Galileo High Accuracy Service (HAS) Decoding Utility · 10-week research internship — Ahmedabad, India

- Built a real-time C++ utility that ingests a multiplexed TCP stream, tokenizes three interleaved protocols (RTCM payloads, Galileo HAS E6-B pages, Time-of-Transfer epoch tags), and routes each to an independent output port.
- Implemented Reed–Solomon (255,32) erasure decoding over GF(2^8) from the Galileo HAS ICD, using lookup-table field arithmetic and Gaussian elimination for k×k matrix inversion, recovering complete messages despite page loss and out-of-order arrival.
- Reverse-engineered four undocumented Galileo RTCM 3.x message formats (1240–1243) and the CRC-24Q framing routine against an open-source implementation and the ICD, which unblocked the output stage.
- Synthesized RTCM 3.x output for GPS PRN 1–32 (messages 1057–1060) and Galileo PRN 1–36 (1240–1243) with correct framing and checksum.
- Delivered TCP streaming over WinSock 2.2 and Win32 serial on a pre-C++11 toolchain that ruled out modern STL containers and threading, using fixed-array structures, background-thread reads and stale-buffer purging.
- Validated end to end in RTKLIB at centimetre-level Precise Point Positioning, unit-testing every decoding stage against a Python reference implementation.

Supervisor: Sri. Pushkar Bhardwaj (Scientist/Engineer). Division Head: Smt. Saumi S. Faculty guide: Prof. Yogesh Trivedi.

Electronics Lead — Sep 2023 – Present
Team Arrow UAV Club, Nirma University · Official aeromodelling and autonomous systems club — Ahmedabad, India

- Lead a team of 8 to 12 across five competition campaigns, owning the full avionics stack from flight-controller bring-up to power design. Four podium finishes, including AIR 2 (SAE DDC Micro 2024), AIR 3 (2025) and 5th internationally on mission score (SAE Aero Design West 2025, USA).
- Trained 13 juniors into subsystem roles: three in piloting, five in autonomous systems and five in fixed-wing electric propulsion.
- Engineered the power plant for the 2025 international entry under a hard 400 W limit, matching motor and propeller on a thrust bench against analytical predictions. Designed the 2026 tilt-rotor VTOL to a 2:1 thrust-to-weight ratio.
- Designed autonomous ground-vehicle payload electronics (magnetometer, time-of-flight ranging, IR line-following) qualifying for the competition's highest 14× capture multiplier. Ran the scoring analysis that set team strategy: autonomous low-payload scores about 52 points against about 30 for manual high-payload.

RESEARCH

- **Simulation and Comparative Analysis of Lightweight Cryptography for Embedded Systems.** Research Methodology & Seminar, Nirma University, 2025–2026. Systematic comparison of AES-128-GCM, DES-CBC, ECC P-256 and ASCON-128 on throughput, peak memory and energy across 5,000 IoT payloads at 32B, 64B and 128B, with significance tested over 10 trial runs.

SELECTED PROJECTS

GF(2^8) Arithmetic Core — Verilog, Exhaustively Verified — 2026
github.com/Kush-Git-379/gf256-rtl

- Rebuilt the Galois-field arithmetic from the ISRO Reed–Solomon erasure decoder as synthesizable RTL over primitive polynomial 0x11D: two GF(2^8) multiplier microarchitectures (log/antilog ROM and combinational shift-and-reduce), table inversion, and multiply-accumulate.
- Wrote self-checking SystemVerilog testbenches driven by golden vectors from the C++ original, covering all 65,536 multiplier pairs and 256 inverse cases. Proved the tests could fail by injecting three seeded faults.

- Quartus synthesis on Cyclone IV E made shift-and-reduce 13× smaller and 1.7× faster than the table version (62 LEs at 7.71 ns against 804 at 12.95 ns). Traced it to asynchronous ROM reads blocking M9K inference; registering the read address recovered block RAM and raised F_{max} to 180 MHz.

RISC-V Cache Locality: Same Instructions, 11.8× Fewer Misses 2026

Modern Processor Architecture (4EC502ME25)

- Wrote seven RV32IM benchmarks plus an RV32IM interpreter and trace-driven cache model in Python (~1,500 lines), then swept 202 cache configurations through them.
- Two source-level changes to a 64×64 matrix multiply, padding the row stride 64→65 words and tiling at TB=4, cut L1 D-cache miss rate from 51.49% to 4.35% (11.8× fewer misses, est. 5.14× speed-up) while issuing the identical 524,288 loads and 4,096 stores.
- A 3C decomposition attributed 205,344 of 272,928 naive-kernel misses to conflicts, showing why associativity cannot fix it and padding can; also found larger blocks make the optimised kernels worse at fixed capacity.

gem5 L1 Cache Sensitivity Analysis Feb 2026

- Swept cache capacity, workload type and set associativity on gem5 v25.1 (X86 TimingSimpleCPU) to separate memory-bound from compute-bound behaviour.
- Growing 4 kB to 64 kB cut matrix-multiply miss rate from 5.06% to 0.04% (IPC +11.9%); 1-way to 8-way cut it from 4.74% to 0.34% (IPC +10.4%). An AES workload held flat at 0.14%, isolating execution units rather than memory as its bottleneck.

Cryptography Benchmark for Resource-Constrained Devices Jan 2026

github.com/Kush-Git-379/crypto-algorithm-benchmark

- Benchmarked AES-GCM, DES, ECC P-256 and ASCON-128 across 5,000 IoT payloads, loading the ASCON C reference as a compiled DLL through Python ctypes for native-speed timing and profiling peak memory with tracemalloc. ASCON ran 78× faster than AES-GCM and 12× faster than DES.

SecureCam — Edge AI Surveillance on Raspberry Pi Apr 2026

Analytics of IoT · team of four

- Built a three-tier surveillance system on a Raspberry Pi 3B: Picamera2 capture over CSI, OpenCV MOG2 motion detection, authenticated MJPEG streaming over HTTP, SQLite session logging, and a Flask REST API serving a live dashboard.
- Exported YOLOv8n to ONNX and profiled it on the Pi; inference was slow enough to stall the video stream, so deep learning stayed on the laptop and the edge node ran MOG2 at 8–15 FPS instead. Concluded that real-time CNN inference on a Pi 3 needs either newer hardware or a dedicated accelerator.
- Sustained 2,256 s of continuous streaming across 39 sessions and 344 motion events at under 500 ms latency, with roughly 60 kB/hour of stored telemetry.

Secure GPS Telemetry Pipeline — ESP32, AES-128, MQTT Mar 2026

github.com/Kush-Git-379/esp32-gps-encryption-tracker

- Built an end-to-end encrypted IoT pipeline: an ESP32 parses NMEA over UART2 with TinyGPS++, serializes to JSON, encrypts with AES-128 through mbedTLS, and publishes ciphertext to an MQTT broker on a 3-second cycle. A Python subscriber decrypts and resolves live position. Verified on hardware from GPS fix to decoded coordinates.

AWARDS & COMPETITIONS

- SAE Aero Design West 2025, Los Angeles, USA — 5th internationally on mission score. Electronics Lead.
- SAE India DDC Micro 2024 — All India Rank 2. Electronics Lead.
- SAE India DDC Micro 2025 — All India Rank 3. Electronics Lead.
- Technoxian 2024, RC Plane Challenge — 4th place.
- SAE Aero Design East 2026, USA and SAE India DDC Micro 2026 — Electronics Lead.

TECHNICAL SKILLS

Languages	C, C++, Verilog HDL, SystemVerilog, Python, Assembly, TCL, Bash, MATLAB, SQL
Digital Design & Verification	RTL design, functional verification, testbench development, golden-reference and exhaustive testing, fault injection, logic synthesis, static timing analysis, ModelSim, Quartus II, FPGA
Computer Architecture	RISC-V RV32IM, gem5, microarchitecture, cache and memory hierarchy, set associativity, IPC and performance analysis, workload characterization, hardware/software co-design
Embedded & Firmware	ESP32, Raspberry Pi, Arduino, 8051, Pixhawk, ArduPilot / Mission Planner, Keil uVision, mbedTLS, bare-metal firmware, device drivers, UART / I²C / SPI, real-time systems
Algorithms & Standards	Reed–Solomon, forward error correction, Galois Field GF(256), Galileo HAS ICD, RTCM 3.x, CRC-24Q, AES-128, ASCON-128, ECC P-256, DSP
Analog & Circuit Design	Cadence Virtuoso, LTspice, SPICE simulation, transient / DC / AC analysis, parametric sweep, pre- and post-layout simulation, MOSFET parameter extraction, current mirrors, CS / CD / CG, cascode and differential amplifier design
Data & ML	pandas, NumPy, Matplotlib, scikit-learn, PyTorch, OpenCV, KNIME, Tableau
Tools	Git, GitHub, Linux, RTKLIB, SolidWorks, Fusion 360, XFLR5